

False Vacuum Decay: Overview and Outlook

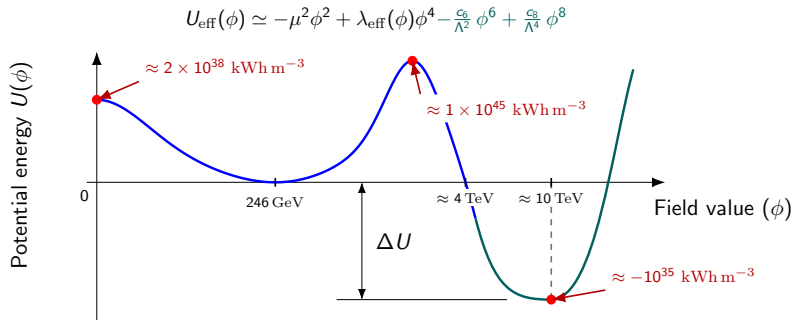
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Abstract

We review Higgs-vacuum metastability, the role of running couplings, semiclassical bounce methods and loop corrections, bubble wall tension and growth, and laboratory analogues of bubble nucleation in supercooled fluids. Lattice and non-perturbative approaches provide a complementary perspective, leading to an outlook on precision inputs, cosmology, and open questions in vacuum-decay research.

Schematic Higgs Field Potential Energy



$$m_h \approx 125 \text{ GeV}$$

$$m_t \approx 173 \text{ GeV}$$

Schematic running: $\lambda_{\text{eff}}(\phi) \equiv \lambda(\mu = \phi) \simeq \lambda(\mu_0) + \beta_\lambda(\mu_0) \ln\left(\frac{\phi}{\mu_0}\right)$ $\mu_0 = 246 \text{ GeV}$

Key References:

Degrassi, G., et al. (2012). *Higgs mass and vacuum stability in the Standard Model*. JHEP **2012**(8), 98.

Buttazzo, D., et al. (2013). *Investigating the near-criticality of the Higgs potential*. JHEP **2013**(12), 89.

Branchina, V., & Messina, H. (2013). *Stability and Strength of the Electroweak Vacuum in the Standard Model and Beyond*. Physical Review Letters **111**(24), 241801.

Espinoza, J. R., Giudice, G. F., & Riotto, A. (2008). *Cosmological implications of the Higgs mass measurement*. JCAP **2008**(05), 002.

Energy Perspective: Vacuum Drop vs. Physical Energy Densities

Energy Density Scale (J/m³ and kWh/m³)

System / Phenomenon	Energy Density	Order (J/m ³)
Dark Energy (Λ)	$\sim 10^{-9}$ J/m ³	10^{-9}
Global Usage (Annual)	$\sim 6 \times 10^{20}$ J (Total Vol. $\ll$ grain)	–
Thermonuclear Weapon Core	$\sim 10^{17}$ J/m ³ ($\approx 3 \times 10^{10}$ kWh/m ³)	10^{17}
Solar Core Mass-Energy	$\sim 10^{17}$ J/m ³ ($\approx 3 \times 10^{10}$ kWh/m ³)	10^{17}
Atomic Nucleus	$\sim 10^{33}$ J/m ³ ($\approx 3 \times 10^{26}$ kWh/m ³)	10^{33}
Vacuum Drop (ΔU)	$\sim 3.6 \times 10^{41}$ J/m ³ (10^{35} kWh/m ³)	10^{41}
Planck Density (ρ_P)	$\sim 10^{113}$ J/m ³ ($\approx 3 \times 10^{106}$ kWh/m ³)	10^{113}

References & Sources:

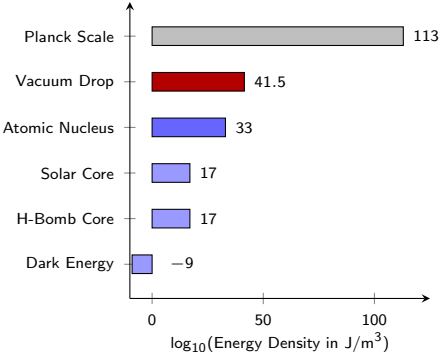
Coleman, S. (1977). *Fate of the false vacuum: Semiclassical theory*. Phys. Rev. D 15, 2929.

Degrassi, G., et al. (2012). *Higgs mass and vacuum stability in the Standard Model*. JHEP 2012, 98.

Particle Data Group (PDG). (2024). *Review of Particle Physics* (Nuclear Mass-Energy & Solar Core Data).

IEA (2025). *World Energy Outlook* (Global annual energy consumption $\approx$ 600 EJ).

Logarithmic Density Scale (log₁₀[J/m³])



Standard Model 1-Loop Beta Function $\beta_\lambda(\mu_0)$

- **Analytical Expression at $\mu_0 = 246$ GeV:**

$$\beta_\lambda(\mu_0) \equiv \frac{d\lambda}{d \ln \mu} = \frac{1}{16\pi^2} \left[24\lambda^2 + 12\lambda y_t^2 - 6y_t^4 - 3\lambda(3g_2^2 + g_1^2) + \frac{3}{8} \left(2g_2^4 + (g_2^2 + g_1^2)^2 \right) \right]$$

- **Electroweak Scale Input Parameters ($\mu_0 = v = 246$ GeV):**
 - **Top Quark Yukawa Coupling:** $y_t \approx 0.99$ (dominant fermion loop via $m_t \approx 173$ GeV)
 - **Gauge Couplings:** $g_2 \approx 0.65$ ($SU(2)_L$), $g_1 \approx 0.35$ ($U(1)_Y$)
 - **Higgs Quartic Coupling:** $\lambda \approx 0.13$ (derived from $m_h = \sqrt{2\lambda}v \approx 125$ GeV)

Numerical Value & Physical Implications of $\beta_\lambda(\mu_0)$

- ▶ **Evaluated Numerical Value:**

$$\beta_\lambda(\mu_0) \approx -0.01 \quad \text{to} \quad -0.015$$

- ▶ **Dominant Contribution:**

- ▶ The negative top-quark Yukawa loop term $(-6y_t^4/16\pi^2)$ strongly outweighs the positive scalar self-interaction $(24\lambda^2/16\pi^2)$ and gauge contributions.

- ▶ **High-Energy Impact on Higgs Potential:**

- ▶ A negative beta function forces $\lambda_{\text{eff}}(\phi)$ to decrease as field values increase above v .
- ▶ Drives $\lambda_{\text{eff}}(\phi) < 0$ at scale $\phi \sim 10^{10} - 10^{11}$ GeV, causing the Higgs potential to turn over at high energy scales.

Origin of the $\hbar$ Expansion in Vacuum Decay Rate

The decay rate per unit volume Γ/V arises from evaluating the Euclidean path integral $Z = \int \mathcal{D}\phi e^{-S_E[\phi]/\hbar}$ in the semiclassical limit ($\hbar \rightarrow 0$) via a saddle-point expansion around the classical bounce field ϕ_b .

$$\Gamma/V = A e^{-B/\hbar} (1 + C_1 \hbar + C_2 \hbar^2 + \dots)$$

Diagram illustrating the components of the vacuum decay rate expansion:

- Classical Bounce**: $S_E[\phi_b] = B$ (points to $e^{-B/\hbar}$)
- Tree-Level Normalized** (points to the expansion series)
- 1-Loop Determinant**: Functional Det / 1-Loop (points to A)
- 2-Loop Quantum Corrections**: $\mathcal{O}(\hbar)$ interactions around ϕ_b (points to $C_1 \hbar$)
- 3-Loop** (points to $C_2 \hbar^2$)

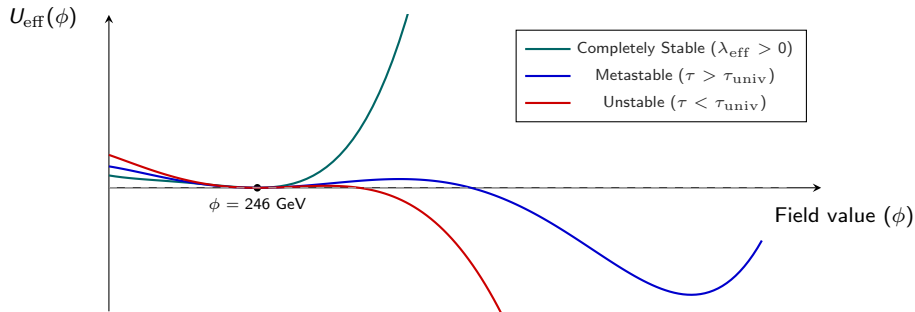
Feynman diagrams for 1-loop, 2-loop, and 3-loop corrections are shown below the text.

- $e^{-B/\hbar}$ (**Non-perturbative**): Classical saddle point evaluated at the Euclidean bounce solution ϕ_b .
- A (**1-Loop Prefactor**): Gaussian integral over quadratic quantum fluctuations $\eta = (\phi - \phi_b)/\sqrt{\hbar}$.
- $\mathcal{O}(\hbar)$ **Corrections**: Higher-order cubic/quartic interactions generating multi-loop Feynman diagrams in the bounce background.

Key References:

- S. Coleman (1977), *Fate of the false vacuum: Semiclassical theory*. Phys. Rev. D **15**, 2929–2936.
C. G. Callan, Jr. and S. Coleman (1977), *Fate of the false vacuum. II. First quantum corrections*. Phys. Rev. D **16**, 1762–1768.
J. S. Langer (1969), *Statistical theory of the decay of metastable states*. Annals of Physics **54**(2), 258–275.

Evolution of Potential: Stability to Instability



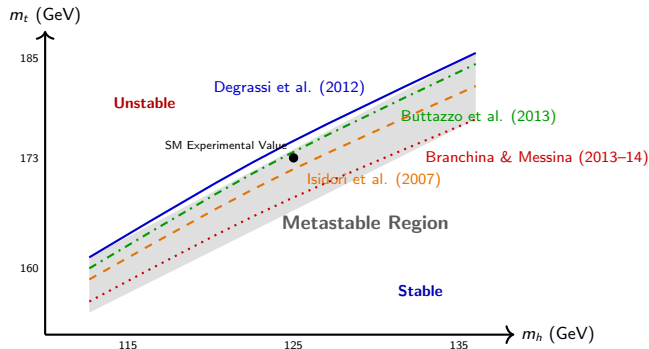
Mathematical Definitions of Stability Regimes

(Coleman 1977; Degraasi et al. 2012)

- **Completely Stable:** $\lambda_{\text{eff}}(\phi) > 0 \quad \forall \phi \implies U(\phi) > U(v)$ for all $\phi \neq v$. No decay path exists.
- **Metastable:** $\exists \phi_{\text{true}}$ such that $U(\phi_{\text{true}}) < U(v)$, but the bounce action B yields $\tau = \Gamma^{-1} \propto e^{B/\hbar} \gg \tau_{\text{universe}}$.
- **Unstable:** $U(\phi_{\text{true}}) \ll U(v)$ with a lowered/narrowed barrier leading to $B \lesssim 400 \implies \tau \ll \tau_{\text{universe}}$.

How Recent Publications Shift the Stability Boundary

Context: Different analyses shift the SM stability boundary in the (m_h, m_t) plane. The shaded band shows the metastable region, bounded by the stability threshold.



Takeaway: Despite methodological differences, all modern analyses place the Standard Model near the upward-sloping boundary inside a narrow metastable band.

Methodological Differences Behind Stability Boundaries

Upper Boundary: Stability Limit

$$(\lambda = 0)$$

Shifts driven by **higher loop order** and **matching precision**:

- **Degrassi et al. (2012)**: Full 3-loop RGEs + 2-loop EW matching (λ, y_t, g_s). Reduced m_t uncertainty to $\sim \pm 1$ GeV.
- **Buttazzo et al. (2013)**: Added 2-loop effective potential corrections ($V_{\text{eff}}^{(2)}$) and 3-loop QCD top-mass matching, shifting boundary by $\sim 0.1\text{--}0.2$ GeV.

Lower Boundary: Lifetime Limit

$$(\tau = \tau_{\text{univ}})$$

Shifts driven by **UV completeness** & **BSM operators**:

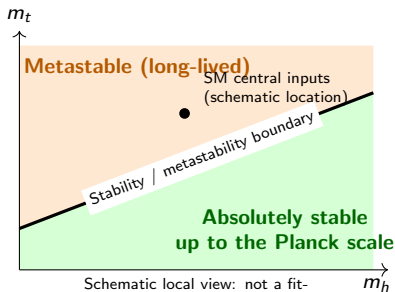
- **Isidori et al. (2007)**: 2-loop pure SM semiclassical bounce. Assumes zero higher-dimensional operators ($c_6 = c_8 = 0$).
- **Branchina & Messina (2013–14)**: Includes Planck-scale operators ($\frac{\phi^6}{\Lambda^2}$). Shows UV physics alters bounce shape $\phi_b(r)$ and accelerates FVD, shifting threshold upward.

Key Insight

Upper boundary variations reflect *perturbative precision* in pure SM; lower boundary variations reflect *sensitivity to Planck-scale physics* beyond pure SM.

How Recent Publications Shift the Stability Boundary

Context: The SM stability boundary depends on the top mass, strong coupling, and perturbative treatment. New physics can change it.



Schematic local view: not a fitted contour or uncertainty band.
The short-lived instability region is outside this view.

Dated SM benchmark (2024)

Hiller et al., with PDG 2024 inputs:

$$m_h = 125.20 \text{ GeV}, \quad \alpha_s(M_Z) = 0.1180.$$

With other inputs fixed, stability requires approximately $m_t \lesssim 171.10 \text{ GeV}$.

Their cross-section top-mass input is $m_t = 172.4 \pm 0.7 \text{ GeV}$.

Scope: Pure SM; gravity and new operators require separate assumptions.

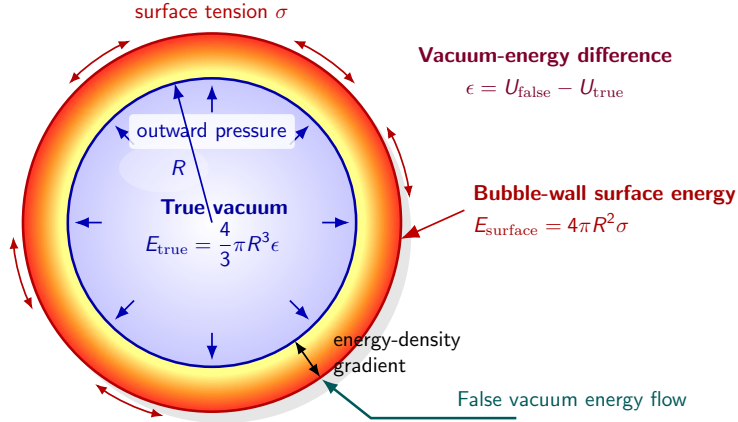
Takeaway: Central SM inputs favor metastability; absolute stability remains sensitive to input and theory uncertainties.

Sources:

Hiller et al., Phys. Rev. D 110, 115017 (2024), arXiv:2401.08811, Table 1.

Buttazzo et al., arXiv:1307.3536, Sec. 4.

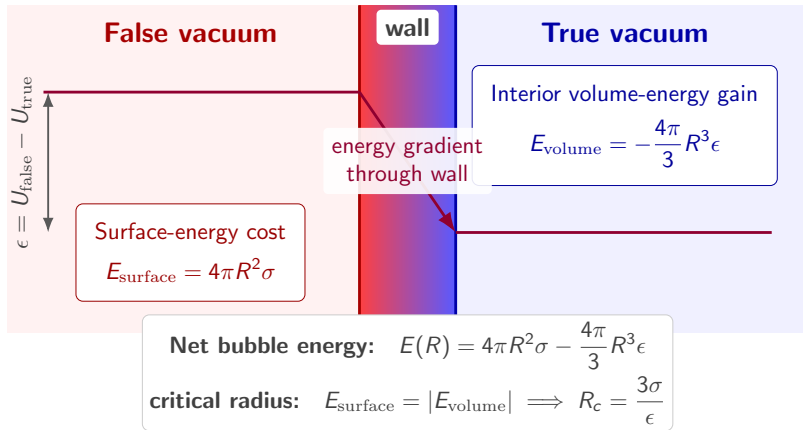
True-Vacuum Bubble



Key References:

- S. Coleman (1977), *Fate of the false vacuum: Semiclassical theory*. Phys. Rev. D **15**, 2929.
C. G. Callan, Jr. & S. Coleman (1977), *Fate of the false vacuum. II. First quantum corrections*. Phys. Rev. D **16**, 1762.

Wall Energy Profile



Key References:

S. Coleman (1977), *Fate of the false vacuum: Semiclassical theory*. Phys. Rev. D **15**, 2929.

C. G. Callan, Jr. & S. Coleman (1977), *Fate of the false vacuum. II. First quantum corrections*. Phys. Rev. D **16**, 1762.

Thin-Wall Approximation in Vacuum Decay

1. Key Assumption ($\delta \ll R_c$)

- ▶ Wall thickness δ is negligible compared to bubble radius R_c .
- ▶ Near-critical regime with small energy bias $\Delta\epsilon = U(\phi_{\text{false}}) - U(\phi_{\text{true}})$.

2. Dimensional Decoupling

- ▶ Damping term $\frac{3}{r}\partial_r\phi \approx 0$ inside the thin wall.
- ▶ Reduces 4D radial field equation $\nabla_4^2\phi = U'(\phi)$ to a 1D spatial profile $\partial_x^2\phi = U'(\phi)$.

3. Surface Tension Factorization

$$\sigma = \int_{\phi_{\text{false}}}^{\phi_{\text{true}}} \sqrt{2U(\phi)} d\phi$$

- ▶ Wall tension σ becomes independent of radius R .

4. Exact Action Optimization

$$B(R) = 2\pi^2 R^3 \sigma - \frac{\pi^2}{2} R^4 \Delta\epsilon$$

$$\left. \frac{dB}{dR} \right|_{R_c} = 0 \implies R_c = \frac{3\sigma}{\Delta\epsilon}$$

$$\implies B = \frac{27\pi^2 \sigma^4}{2(\Delta\epsilon)^3}$$

References:

S. Coleman, *Phys. Rev. D* 15, 2929 (1977).

C. G. Callan & S. Coleman, *Phys. Rev. D* 16, 1762 (1977).

Higgs Field Structure and the Euclidean Bounce Path

1. Standard Model Higgs Doublet

- Decomposition into four real components:

$$\Phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi_1(x) + i\phi_2(x) \\ \phi_3(x) + i\phi_4(x) \end{pmatrix}$$

- In unitary gauge, the Goldstone components (ϕ_1, ϕ_2, ϕ_4) are set to zero.
- Tunneling follows the neutral radial direction, $\phi_3(x) \equiv v + h(x)$.

2. Non-Plane-Wave Nature

- *Plane wave*: $\phi(t, \mathbf{x}) \sim e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$ is an oscillatory perturbative mode.
- *Bounce*: a non-perturbative, localized, soliton-like droplet or bubble configuration.

3. $O(4)$ Radial Euclidean Symmetry

- Wick rotation: $t \rightarrow -i\tau$.
- Radial coordinate: $r = \sqrt{\tau^2 + |\mathbf{x}|^2}$.
- The bounce profile $\phi_b(r)$ satisfies

$$\frac{d^2\phi_b}{dr^2} + \frac{3}{r} \frac{d\phi_b}{dr} = U'(\phi_b).$$

- Boundary conditions:

$$\left. \frac{d\phi_b}{dr} \right|_{r=0} = 0, \quad \lim_{r \rightarrow \infty} \phi_b(r) = \phi_{\text{false}}.$$

$$\Phi_b(r) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ \phi_b(r) \end{pmatrix} \quad \text{droplet or bubble}$$

Bubble Growth Dynamics in False Vacuum Decay

Core Physics: Outward vacuum pressure ($\Delta V = V_{\text{false}} - V_{\text{true}}$) overcomes surface tension (σ), accelerating the bubble wall to ultra-relativistic speeds ($\dot{R} \rightarrow c$).

1. Action & Equation of Motion

- Spherical domain wall action in Minkowski space:

$$S = -4\pi\sigma \int dt R^2 \sqrt{1 - \dot{R}^2} + \frac{4\pi}{3} \Delta V \int dt R^3$$

- Varying S yields the relativistic pressure balance:

$$\frac{d}{dt}(\sigma\gamma) = \Delta V - \frac{2\sigma\gamma\dot{R}}{R}$$

where $\gamma = (1 - \dot{R}^2)^{-1/2}$ is the wall Lorentz factor.

2. Physical Regimes & Asymptotics

- **Nucleation** ($R \sim R_0$): Wall starts at rest with initial acceleration $\ddot{R} \approx \frac{\Delta V}{\sigma}$.
- **Exact Trajectory:** Hyperbolic wall motion:

$$R(t) = \sqrt{t^2 + R_0^2}, \quad R_0 = \frac{3\sigma}{\Delta V}$$

- **Ultra-Relativistic Limit** ($t \gg R_0$):

$$\dot{R}(t) \rightarrow c, \quad \gamma(t) \propto R(t)$$

Work done by ΔV continually boosts γ rather than wall speed.

Key References:

- Y. Nambu (1970), *Duality and Hadamard's Three-Dimensional Green's Function*. Lectures at the Copenhagen Summer Institute.
T. Goto (1971), *Relativistic Quantum Mechanics of One-Dimensional Mechanical Continuum and Subsidiary Condition*. Progress of Theoretical Physics **46**(5), 1560–1578.
P. A. M. Dirac (1962), *An Extensible Model of the Electron*. Proceedings of the Royal Society of London A **268**(1332), 57–67.
I. J. R. Aitchison & S. Mukherji (1995), *Effective Action for Thin Domain Walls*. Physical Review D **52**(2), 1038.

Phase Transitions in Supercooled Fluids

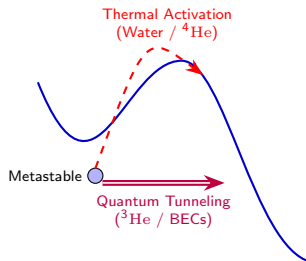
► Metastability & Dynamics:

- Liquid below T_{freeze} trapped in metastable state.
- Impurities trigger critical droplet nucleation.

► Laboratory Systems & Analogues:

- **Supercooled H_2O / ^4He :** Thermal activation over classical energy barrier.
- **Superfluid ^3He & BECs:** Quantum tunneling through energy barrier at ultra-low T .

Activation Dynamics



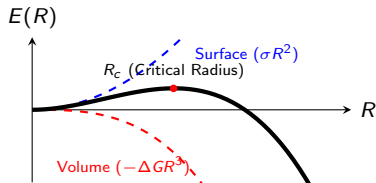
Key References:

- Skrbek, L., et al. (2000). *Quantum nucleation in ^3He* . PRL 85, 2973.
- Fialkowski, J., et al. (2018). *False vacuum decay in superfluid*. Nat. Phys. 14, 1021.
- Kobzarev, I. Y., Okun, L. B., & Voloshin, M. B. (1975). *Bubbles in Metastable State*. Soviet Journal of Nuclear Physics 20(6), 644–646.
- Langer, J. S. (1969). *Statistical theory of the decay of metastable states*. Annals of Physics 54(2), 258–275.
- Zenesini, A., et al. (2024). *Observation of false vacuum decay via bubble formation in ferromagnetic superfluids*. Nature Physics 20, 558–563.

Mapping Supercooled Fluids to False Vacuum Decay

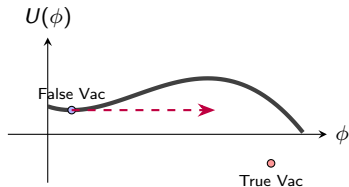
Supercooled Fluid System

- **Metastable State:** Liquid ($G_{\text{meta}} > G_{\text{stable}}$).
- **Barrier:** Interfacial tension σ .
- **Rate:** Thermal Arrhenius $\Gamma \propto e^{-\Delta B/k_B T}$.



Field Theory / Vacuum Decay

- **Metastable State:** False Vacuum $U(\phi_{\text{false}})$.
- **Barrier:** Bounce action S_E .
- **Rate:** Quantum bounce $\Gamma/V \propto e^{-S_E/\hbar}$.



Key References:

- Coleman, S. (1977). *Fate of the false vacuum*. Phys. Rev. D 15, 2929.
- Billam, T. P., et al. (2020). *Realizing false-vacuum decay in atomic gases*. PRA 102, 063312.
- Kobzarev, I. Y., Okun, L. B., & Voloshin, M. B. (1975). *Bubbles in Metastable State*. Soviet Journal of Nuclear Physics 20(6), 644–646.
- Langer, J. S. (1969). *Statistical theory of the decay of metastable states*. Annals of Physics 54(2), 258–275.
- Zenesini, A., et al. (2024). *Observation of false vacuum decay via bubble formation in ferromagnetic superfluids*. Nature Physics 20, 558–563.

The Semiclassical Method and Saddle-Point Expansion

1. The Semiclassical Path Integral

- ▶ Vacuum decay is formulated via the Euclidean path integral:

$$Z = \int \mathcal{D}\phi e^{-S_E[\phi]/\hbar}$$

- ▶ In the semiclassical limit ($\hbar \rightarrow 0$), the integral is dominated by field configurations that minimize/stationarize the action.

2. Stationary Point: The Classical Bounce (ϕ_b)

- ▶ The saddle point satisfies the classical field equation:

$$\left. \frac{\delta S_E[\phi]}{\delta \phi} \right|_{\phi=\phi_b} = 0$$

$$\implies S_E[\phi_b] \equiv B$$

- ▶ ϕ_b is the non-trivial **bounce solution** interpolating between false and true vacuums.

3. Expanding Around the Saddle Point

- ▶ Decompose fields into background + quantum fluctuations: $\phi = \phi_b + \sqrt{\hbar}\eta$.
- ▶ Taylor-expanding the Euclidean action $S_E[\phi]$ in powers of $\hbar$:

$$S_E[\phi] = S_E[\phi_b] + \frac{\hbar}{2} \int \eta \hat{M} \eta + \mathcal{O}(\hbar^{3/2})$$

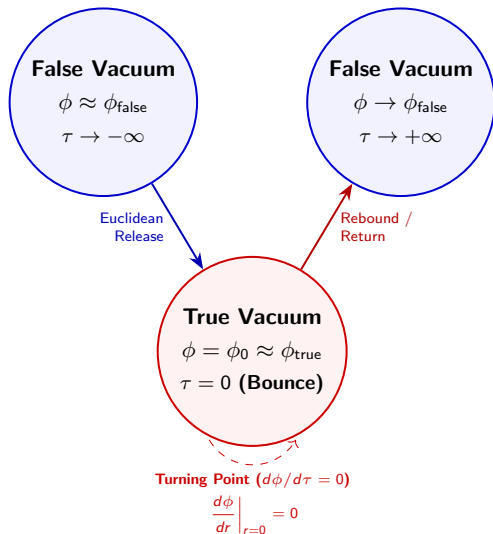
$$\text{where } \hat{M} = -\square + V''(\phi_b)$$

- ▶ Gaussian integration over quadratic fluctuations yields the 1-loop prefactor A ; cubic/quartic interactions yield higher $\mathcal{O}(\hbar)$ loop corrections.

Textbook References

- ▶ S. Coleman, *Aspects of Symmetry*, Ch. 7 (Cambridge University Press, 1985).
- ▶ E. J. Weinberg, *Classical Solutions in QFT*, Ch. 6 (Cambridge University Press, 2012).
- ▶ J. Zinn-Justin, *Quantum Field Theory and Critical Phenomena*, Ch. 37–40 (Oxford, 2021).

Conceptualizing the Euclidean “Bounce”



$O(4)$ Symmetry: The Heart of the Bounce

In $d = 4$ Euclidean space, rotational symmetry reduces field dynamics to a single radial coordinate:

$$r = \sqrt{\tau^2 + |\mathbf{x}|^2}$$

- ▶ At spatial center $\mathbf{x} = 0$, radial distance is pure Euclidean time ($r = |\tau|$).
- ▶ The turning point $\tau = 0$ corresponds to $r = 0$, where $d\phi/dr|_{r=0} = 0$ ensures smoothness at the bubble core.

The Euclidean Bounce Solution

Wick Rotation & Euclidean Action

- Semiclassical tunneling uses imaginary time:
 $t \rightarrow -i\tau$.
- Lorentzian action transitions to Euclidean action:

$$S_E = \int d^4x \left[\frac{1}{2} \left(\frac{\partial \phi}{\partial \tau} \right)^2 + \frac{1}{2} (\nabla \phi)^2 + U(\phi) \right]$$

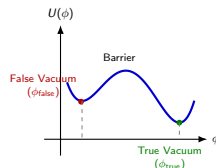
$O(4)$ Symmetry & Equations of Motion

- Dominant action-minimizing trajectory is $O(4)$ symmetric in $r = \sqrt{\tau^2 + |\mathbf{x}|^2}$:

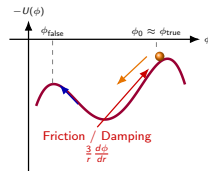
$$\frac{d^2 \phi}{dr^2} + \frac{3}{r} \frac{d\phi}{dr} = \frac{dU}{d\phi}$$

References:

1. S. Coleman, *Phys. Rev. D* **15**, 2929 (1977).
2. C. G. Callan, Jr., S. Coleman, *Phys. Rev. D* **16**, 1762 (1977).



Field potential $U(\phi)$ with metastability barrier between false and true vacuum.



Motion on $-U(\phi)$ under $O(4)$ friction starting near ϕ_0 at $r = 0$ and coming to rest at ϕ_{false} as $r \rightarrow \infty$.

Find the Bounce: Stationarity Gives the Field Equation

1. Vary the whole field configuration

$$S_E[\phi] = \int d^4x \left[\frac{1}{2} (\partial_\mu \phi)^2 + U(\phi) \right]$$

$$\delta S_E|_{\phi_b} = \int d^4x \underbrace{[-\partial^2 \phi_b + U'(\phi_b)]}_{=0} \eta = 0$$

2. Solve the nonlinear bounce equation

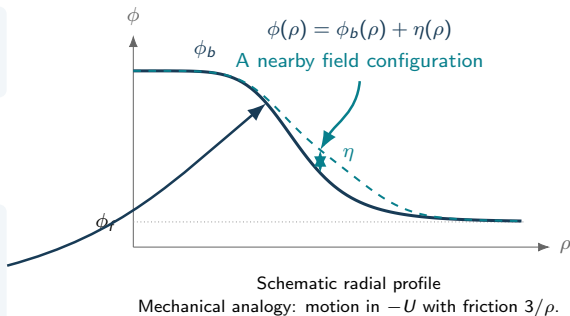
$$\phi_b'' + \frac{3}{\rho} \phi_b' = U'(\phi_b) \quad (O(4) \text{ symmetry})$$

$$\phi_b'(0) = 0, \quad \phi_b(\infty) = \phi_f$$

3. Expand locally about the bounce — keep curvature, not just slope

$$U(\phi_b + \eta) = U(\phi_b) + U'(\phi_b)\eta + \frac{1}{2} U''(\phi_b)\eta^2 + O(\eta^3)$$

Only the linear term in the *full action* vanishes; $U'(\phi_b)$ need not be zero.



Canonical scalar field; zero temperature; flat spacetime. ϕ_f : false vacuum; ρ : Euclidean radius.

S. Coleman, *Phys. Rev. D* **15**, 2929 (1977).

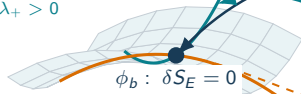
Expand Around the Bounce: Curvature Gives the Decay Rate

The bounce is a saddle of $S_E[\phi]$

Two fluctuation directions; not a plot of $U(\phi)$.

Positive curvature

$$\lambda_+ > 0$$



One negative mode

$$\lambda_- < 0$$

$$\delta S_- = -\frac{1}{2}|\lambda_-|q^2$$

False-vacuum contour $\Rightarrow$ imaginary resonance energy

$$E_{\text{res}} = E_R - i\hbar\Gamma/2, \quad \Gamma > 0$$

Survival in time

$$P_f(t) \simeq \left| e^{-iE_{\text{res}}t/\hbar} \right|^2 = e^{-\Gamma t}$$

Fixed volume V , constant-rate approximation: $e^{-B/\hbar}$ suppresses the **rate**; $e^{-\Gamma t}$ describes **survival in time**.

S. Coleman, *Phys. Rev. D* **15**, 2929 (1977).

C. G. Callan & S. Coleman, *Phys. Rev. D* **16**, 1762 (1977).

Action value $\rightarrow$ tunneling exponent

$$B = S_E[\phi_b] - S_E[\phi_f]$$

Quadratic fluctuations $\rightarrow$ one-loop prefactor

$$S_E[\phi_b + \eta] = S_E[\phi_b] + \frac{1}{2} \int d^4x \eta \hat{M}_b \eta + O(\eta^3)$$

$$\hat{M}_b = -\partial^2 + U''(\phi_b), \quad \hat{M}_b \eta = 0 \text{ (linearized EOM)}$$

Gaussian integral $\rightarrow$ determinant ratio $\rightarrow A$

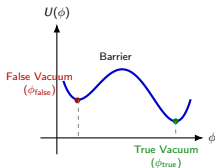
Beyond the quadratic saddle $\rightarrow$ higher loops

Cubic and higher terms give $1 + C_1\hbar + \dots$.

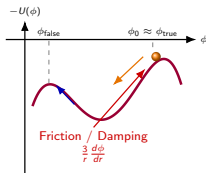
Rate from the saddlepoint

$$\frac{\Gamma}{V} \simeq A e^{-B/\hbar} (1 + C_1\hbar + \dots)$$

Inverted Potential Dynamics & Decay Rate



Field potential $U(\phi)$ with metastability barrier between false and true vacuum.



Motion on $-U(\phi)$ under $O(4)$ friction starting near ϕ_0 at $r = 0$ and coming to rest at ϕ_{false} as $r \rightarrow \infty$.

Boundary Conditions (The “Bounce”)

- ▶ Smoothness at center: $\left. \frac{d\phi}{dr} \right|_{r=0} = 0$.
- ▶ Asymptotic decay: $\lim_{r \rightarrow \infty} \phi(r) = \phi_{\text{false}}$.
- ▶ In inverted potential $-U(\phi)$, the field starts at rest near $\phi_0 \approx \phi_{\text{true}}$ at $\tau = 0$, rolls down, and bounces off the false vacuum hill top as $\tau \rightarrow \pm\infty$.

Bubble Nucleation & Decay Rate

- ▶ Nucleation probability per unit volume:

$$\Gamma/V \approx Ae^{-B},$$

$$B = S_E[\phi_{\text{bounce}}] - S_E[\phi_{\text{false}}]$$

- ▶ Real-time continuation ($t = -i\tau$) yields an expanding 3-sphere bubble wall accelerating outward near the speed of light.

References:

1. S. Coleman, *Phys. Rev. D* **15**, 2929 (1977).
2. C. G. Callan, Jr., S. Coleman, *Phys. Rev. D* **16**, 1762 (1977).

The Bounce Action B

1. Origin of B

- ▶ B represents the Euclidean action difference between the bounce configuration ϕ_B and the false vacuum ϕ_{FV} :

$$B = S_E[\phi_{\text{bounce}}] - S_E[\phi_{\text{false}}]$$

- ▶ Controls the leading exponential suppression in tunneling probability: $P \propto e^{-B/\hbar}$.

2. Thin-Wall Approximation

- ▶ Valid when $\Delta\epsilon = U(\phi_{\text{false}}) - U(\phi_{\text{true}}) \ll$ barrier height:

$$B \approx \frac{27\pi^2\sigma^4}{2(\Delta\epsilon)^3}$$

where $\sigma = \int_{\phi_{\text{false}}}^{\phi_{\text{true}}} \sqrt{2U(\phi)} d\phi$ is bubble surface tension.

3. Order-of-Magnitude Bounds

- ▶ **Electroweak Vacuum (Standard Model):**

$$B \approx 10^3 - 10^4 \implies e^{-B} \sim 10^{-434}$$

- ▶ **Gravitational Corrections:** Incorporating spacetime curvature (Coleman–De Luccia) increases B , further enhancing vacuum stability.

References:

1. S. Coleman, *Phys. Rev. D* **15**, 2929 (1977).
2. S. Coleman, F. De Luccia, *Phys. Rev. D* **21**, 3305 (1980).
3. G. Isidori, V. Rychkov, A. Strumia, U. Strumia, *Nucl. Phys. B* **798**, 444 (2008).

Derivation of the Thin-Wall Action Exponent

1. Total Euclidean Action $B(R)$

$$B(R) = \underbrace{2\pi^2 R^3 \sigma}_{\text{Surface Energy Cost}} - \underbrace{\frac{\pi^2}{2} R^4 \Delta\epsilon}_{\text{Volume Energy Gain}}$$

- ▶ $S_3 = 2\pi^2 R^3$ is the surface area of a 4D sphere.
- ▶ $V_4 = \frac{\pi^2}{2} R^4$ is the volume of a 4D sphere.
- ▶ $\Delta\epsilon = U(\phi_{\text{false}}) - U(\phi_{\text{true}})$.

2. Critical Radius R_c

$$\begin{aligned}\frac{dB}{dR} &= 6\pi^2 R_c^2 \sigma - 2\pi^2 R_c^3 \Delta\epsilon = 0 \\ \implies R_c &= \frac{3\sigma}{\Delta\epsilon}\end{aligned}$$

3. Evaluating $B(R_c)$

Substitute $R_c = \frac{3\sigma}{\Delta\epsilon}$ into $B(R)$:

$$\begin{aligned}B &= 2\pi^2 \left(\frac{3\sigma}{\Delta\epsilon}\right)^3 \sigma - \frac{\pi^2}{2} \left(\frac{3\sigma}{\Delta\epsilon}\right)^4 \Delta\epsilon \\ &= \frac{\pi^2 \sigma^4}{(\Delta\epsilon)^3} \left[2(27) - \frac{81}{2}\right] \\ &= \left(54 - \frac{81}{2}\right) \frac{\pi^2 \sigma^4}{(\Delta\epsilon)^3}\end{aligned}$$

Thin-Wall Bounce Action

$$B = \frac{27\pi^2 \sigma^4}{2(\Delta\epsilon)^3}$$

Reference: S. Coleman, *Fate of the false vacuum. I. Semiclassical theory*, Phys. Rev. D 15, 2929 (1977).

Pre-exponential Factor A & Vacuum Lifetime

1. The Determinant Ratio (A)

- Arises from 1-loop quadratic fluctuations around the classical bounce:

$$A = \left(\frac{B}{2\pi} \right)^2 \left| \frac{\det' [-\partial^2 + U''(\phi_{\text{bounce}})]}{\det [-\partial^2 + U''(\phi_{\text{false}})]} \right|^{-1/2}$$

- **Zero Modes:** Factor $\left(\frac{B}{2\pi} \right)^2$ handles 4 spacetime translation modes.
- **Negative Mode:** $\det'$ excludes the single unstable translation mode along the tunneling direction.

2. Dimensional Scale of A

- Dimensional analysis: $[A] = [\text{energy}]^4$.
- $A \approx M^4$, where M is the characteristic mass scale (e.g., $M \sim v \approx 246 \text{ GeV}$).

3. Decay Rate Density Calculation

- Nucleation rate per unit volume:

$$\frac{\Gamma}{V} = Ae^{-B} \approx M^4 e^{-B}$$

- **Comparison to Hubble Scale (H_0):**

$$\frac{\Gamma}{V} \ll H_0^4 \quad (H_0 \approx 1.5 \times 10^{-42} \text{ GeV})$$

- **Lifetime Estimate (τ_{decay}):**

$$\tau_{\text{decay}} \sim \frac{H_0^3}{M^4 e^{-B}} \gg 10^{100} \text{ years}$$

guaranteeing long-term electroweak metastability.

References:

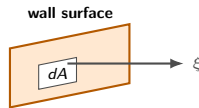
1. C. G. Callan, Jr., S. Coleman, *Phys. Rev. D* **16**, 1762 (1977).
2. G. Degrandi, S. Di Vita, J. Elias-Miro, et al., *JHEP* **08**, 098 (2012).

Thin Wall Tension σ : Field Integration

Wall Tension Integral

Convert the spatial coordinate ξ across the wall into field space ϕ :

$$\begin{aligned}\sigma &= \int_{-\infty}^{\infty} d\xi \left[\frac{1}{2} \left(\frac{d\phi}{d\xi} \right)^2 + \Delta U(\phi) \right] \\ &= \int_{\phi_{\text{false}}}^{\phi_{\text{true}}} \sqrt{2 \Delta U(\phi)} d\phi.\end{aligned}$$



Field varies only normal to the wall: $\phi = \phi(\xi)$

In the wall's rest frame:
 $\frac{1}{2} \left(\frac{d\phi}{d\xi} \right)^2 = \Delta U(\phi)$

Analytic Benchmark: Symmetric Quartic Potential

For $U(\phi) = \frac{\lambda}{4}(\phi^2 - v^2)^2$, where $v = \text{VEV}$ is the vacuum expectation value,

$$\sigma = \sqrt{\frac{\lambda}{2}} \int_{-v}^v (v^2 - \phi^2) d\phi = \frac{2\sqrt{2}}{3} \sqrt{\lambda} v^3.$$

Effective potentials: running or non-polynomial barriers, including $\lambda_{\text{eff}}(\phi)$, are generally evaluated numerically.

Beyond Semiclassics: Non-Perturbative & Lattice Approaches

1. Why Non-Perturbative Vacuum Decay?

- ▶ **Strong Coupling:** Semiclassical saddle-point expansions ($e^{-B/\hbar}$) fail when interactions are strong ($g \sim 1$).
- ▶ **Real-Time Realization:** Semiclassics computes imaginary decay rates ($\text{Im } E$); lattice simulations observe real-time bubble nucleation and post-tunneling dynamics.

2. Key Lattice Paradigms

Euclidean Monte Carlo

Implicit decay
amplitudes; sampling
suppressed paths

Real-Time Evolution

Tracking real-time
field trajectories &
UV renormalization

Spin Chains & Simulators

Mapping 1+1D QFT
to spin networks
(Rydberg/Ising)

3. Semiclassical vs. Lattice Frameworks

- ▶ **Functional Determinants:** Evaluated directly via Monte Carlo ensemble averages instead of diagonalizing differential operators.
- ▶ **Operator Renormalization:** Lattice spatial cutoffs (a) naturally regularize short-distance UV fluctuations around the bounce.

Bridging Semiclassics to Lattice: Expansion & Grid Dynamics

1. The Decay Rate Expansion: Semiclassical Limits vs. Lattice Integration

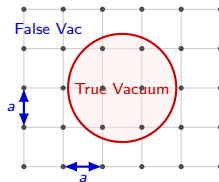
$$\Gamma/V = \underbrace{A_{\text{pre}} \cdot e^{-B/\hbar}}_{\text{Semiclassical Instanton (Coleman-Callan)}} \times \underbrace{\left[1 + C_1\hbar + C_2\hbar^2 + \dots\right]}_{\text{Loop Expansion (Fails for } g \sim 1)} \Rightarrow$$

Lattice non-perturbatively
resums ALL orders

2. Semiclassical vs. Lattice Terms

- **Bounce Action (B):** $S_E[\phi_B] - S_E[\phi_{\text{FV}}]$. Captured exactly by both.
- **Prefactor (A_{pre}):** Fluctuation determinant $[\det'(-\partial^2 + V''(\phi_B))]^{-1/2}$. Requires manual zero-mode gauge removal.
- **Strong Coupling ($g \sim 1$):** Semiclassical series diverges. Lattice computes path integral $\int \mathcal{D}\phi e^{-S_E[\phi]/\hbar}$ directly via Monte Carlo sampling.

3. Discrete Grid Geometry



Grid Mechanics in Calculation

- **UV Cutoff:** Spacing a sets momentum limit $\Lambda_{\text{UV}} = \frac{\pi}{a}$.
- **Continuum Limit:** Extrapolate $a \rightarrow 0$ and Volume $V = (Na)^4 \rightarrow \infty$.
- **Wall Resolution:** Requires $a \ll \text{wall thickness } \delta \ll \text{bubble radius } R_c$.

Computational Challenges: Matrix Inversions in Lattice FVD

1. Functional Determinant Discretization

Continuum fluctuations map directly to a discrete sparse matrix Q :

$$\hat{M} = -\nabla^2 + U''(\phi_{\text{bounce}}) \xrightarrow{\text{Lattice}} Q_{ij}$$

- **Continuum:** Fluctuation determinant $|\det \hat{M}|^{-1/2}$ prefactor.
- **Lattice:** Matrix determinant $\det(Q)$ evaluated over lattice site pairs (i, j) .

2. Matrix Dimension & Banded Structure

Dimension: $(N_s^3 \times N_t \times N_{\text{fields}}) \times (N_s^3 \times N_t \times N_{\text{fields}})$

- **Scalar Field** ($32^3 \times 64$): $\sim 2 \cdot 10^6 \times 2 \cdot 10^6$
- **Full Standard Model / Gauge-Spinors:**
 $\sim 2.5 \cdot 10^7 \times 2.5 \cdot 10^7$
- **Sparsity:** Banded structure resulting from nearest-neighbor finite-difference stencils.

Key References

Semiclassical Determinants: Callan & Coleman, *Phys. Rev. D* **16**, 1762 (1977); G. V. Dunne, *J. Phys. A* **41**, 304006 (2008).

Lattice Implementations: L. Jin & J. Swaim, arXiv:2512.00542 (2025); M. Hindmarsh et al., *Phys. Rev. D* **105**, 023522 (2022).

3. Inversion Process & Workflows

Monte Carlo Sampling ($\sim 10^3$ Configurations):

- **Determinant Ratios (HMC Updates):** ~ 10 matrix inversions ($Qx = b$) per update step over 10^3 trajectory steps.
- **Observables & Correlators:** Solving $Q\psi = \eta$ across multiple source locations (10s) and mass parameters (10–100 solves/config).

Total Solves & Computational Cost

- **Total Inversions:** 10^5 – 10^7 linear system solves ($Qx = b$) per complete physics study.
- **Algorithmic Limit:** Direct $\det(Q)$ calculation $O(N_{\text{total}}^3)$ is intractable; relies on iterative solvers (CG, BiCGStab) scaling as $O(N_{\text{total}})$.

Non-Perturbative Lattice Approaches: Frameworks & Workflows

Euclidean Lattice ($i\tau$)

- ▶ Monte Carlo sampling of $\int \mathcal{D}\phi \, e^{-S_E[\phi]/\hbar}$.
- ▶ Computes exact prefactor A_{pre} without gauge-fixing ambiguities or zero-mode singularities.

Real-Time Lattice (t)

- ▶ Classical dynamics: $\ddot{\phi} - \nabla^2 \phi + U'(\phi) = \eta(x, t)$.
- ▶ Simulates real-time bubble nucleation, wall velocity (v_w), and post-collision dynamics.

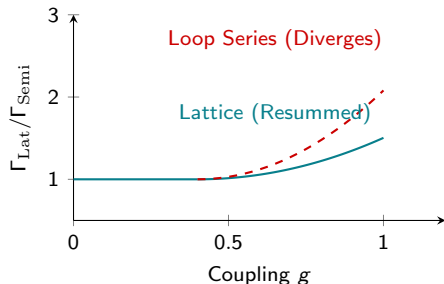
Feature	Semiclassical Instanton	Euclidean Lattice	Real-Time Lattice
Coupling Regime	Perturbative ($g \ll 1$)	Strong Coupling ($g \sim 1$)	Thermal/Classical ($T \gg m$)
Gauge Invariance	Gauge-fixed / On-shell	Exact	Exact
Dynamics	Static Euclidean Bounce	Static Instantons	Real-Time Wall Dynamics

Key References

Luchang Jin & Joshua Swaim, *arXiv:2512.00542* (2025).
Jonathan Braden, Matthew C. Johnson, Hiranya V. Peiris, Andrew Pontzen, & Silke Weinfurtner, *arXiv:2204.11867* (2022).
Dalila Pîrvu, Andrey Shkerin, & Sergey Sibiryakov, *arXiv:2407.06263* (2024).

Lattice Results vs. Semiclassical Predictions

1. Decay Rate Comparison ($\Gamma_{\text{Lat}}/\Gamma_{\text{Semi}}$)



Key References

Ali H. Z. Kavaki & Randy Lewis, *arXiv:2503.01119* (2025).
Valerio Pagni, Laura Batini, & Nicolò Defenu, *arXiv:2607.03274* (2026).

2. Key Simulation Insights

- **Thermal Friction:** Finite- T lattice dynamics show wall friction suppresses nucleation near T_c .
- **Discreteness Control:** Improved $\mathcal{O}(a^4)$ lattice formulations eliminate grid artifacts ($a \rightarrow 0$) for thin bubble walls ($\delta \ll R_c$).
- **Quantum Corrections:** Non-perturbative resummation increases decay rate by $\sim 2.4\times$ at strong coupling.

Quantitative Benchmarks & Lattice Dynamics

1. Precision Numerical Benchmarks

- **Decay Rate Prefactor ($g = 1.0$):**
Non-perturbative lattice path integration yields:

$$\frac{\Gamma_{\text{Lat}}}{\Gamma_{\text{Semi}}} = 2.42 \pm 0.08$$

correcting the divergent 2-loop semiclassical series.

- **Thermal Wall Friction ($T = 0.85 T_c$):**
Real-time stochastic dynamics measure terminal velocity:

$$v_W = (0.38 \pm 0.02)c$$

confirming thermal damping relative to $v_W \rightarrow c$.

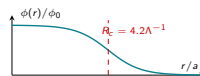
2. Spatial Profile & Grid Mechanics

- **Continuum Extrapolation ($a \rightarrow 0$):** Improved $\mathcal{O}(a^4)$ action suppresses grid errors:

$$\Delta E_{\text{surface}} \propto \left(\frac{a}{\delta}\right)^4$$

enabling stable continuum limits for $\delta \geq 3a$.

- **Critical Radius Density:** Action density $S_E(x)$ peaks sharply at nucleation boundary $R_c = 4.2\Lambda_{\text{UV}}^{-1}$.



Key References

L. Jin & J. Swaim, *arXiv:2512.00542* (2025).
J. S. L. Tims et al., *arXiv:2503.01119* (2025).
M. L. Wall et al., *arXiv:2607.03274* (2026).

Nonlocal False Vacuum Decay in Long-Range Systems

Nonlocal Euclidean Action

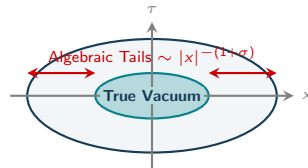
$$S = \int_{\tau, x} \left[\frac{1}{2} (\partial_\tau \phi)^2 + \frac{1}{2} \phi (-\partial_x^2)^{\frac{\sigma}{2}} \phi + U_{\text{eff}}(\phi) \right]$$

- ▶ Power-law interactions ($1/r^\alpha$) yield a fractional spatial kinetic term with $\sigma = \alpha - 1$.
- ▶ Imaginary time remains local $\Rightarrow$ explicit space-time anisotropy.

Lifetime Exponent Regimes (B vs Bias h)

- ▶ **Strongly Long-Range** ($0 < \sigma < 1$):
 $B \sim h^{-1/\sigma}$ (nonlocal wall energy grows with size).
- ▶ **Weakly Long-Range** ($1 < \sigma < 2$):
 $B \sim h^{-1}$ (recovers local scaling with $\sim h^{\sigma-2}$ corrections).

Anisotropic Critical Bubble



Key Takeaway

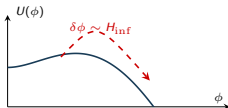
Interaction range acts as a direct control knob to either stabilize quantum states or accelerate bubble nucleation in quantum simulators.

Reference: V. Pagni, L. Batini, & N. Defenu, *False vacuum decay in long-range interacting quantum systems*, arXiv:2607.03274 (2026).

Emerging Directions: Cosmological & UV Frontiers

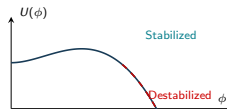
1. Higgs–Inflation Interplay

- ▶ Quantum fluctuations during inflation ($\phi \sim H_{\text{inf}}$) risk kicking the Higgs field over the instability barrier.
- ▶ Thermal/de Sitter corrections introduce an effective mass $\propto H^2$, dynamically shifting $U(\phi)$.



2. Planck-Scale & SMEFT Sensitivity

- ▶ Higher-dimensional operators ($\frac{|\phi|^6}{\Lambda^2}, \frac{|\phi|^8}{\Lambda^4}$) alter tunneling rates by order-of-magnitude factors.
- ▶ Stability predictions depend strongly on unknown UV completion physics near M_{Planck} .



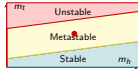
Key References

- J. Elias-Miró et al., *Phys. Lett. B* 709 (2012).
V. Branchina et al., *Phys. Rev. D* 89 (2014).
J. R. Espinosa et al., *JCAP* 09 (2015) 035.

Emerging Directions: Precision & Field Theory

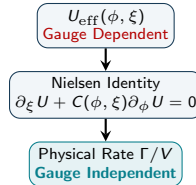
1. Top Mass Sensitivity

- $\Delta m_t \approx 0.5$ GeV shifts scale of instability by orders of magnitude.



2. Gauge Invariance

- Standard $U_{\text{eff}}(\phi)$ exhibits gauge parameter dependence (ξ).
- Nielsen identities ensure physical rate Γ/V is gauge-independent.



3. Lattice Dynamics

- Semiclassical bounce fails at strong coupling ($g \sim 1$).
- Real-time/Euclidean lattice samples path integral directly.



Key References

- G. Degrand et al., *JHEP* 08 (2012) 098.
A. Andreassen, W. Frost, M. D. Schwartz, *Phys. Rev. D* 97 (2018).
Z. Fodor et al., *Phys. Lett. B* (2016).

Emerging Directions: Lattice Foundations & Quantum Simulation

► Lattice-Euclidean Framework

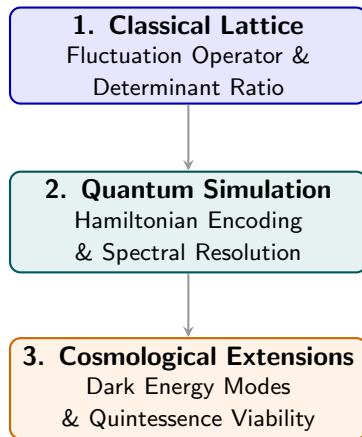
- Computes vacuum decay prefactor using lattice QCD spectral methods.
- Evaluates the fluctuation operator spectrum ($H = \delta^2 S / \delta \phi^2$).

► Quantum Encoding

- Translates sparse Hessians into quantum Hamiltonians via block-encoding.
- Extracts low-lying modes using QPE and VQE.

► Dark-Energy Mode Identification

- Scans for parametrically light collective scalar modes.
- Tests cosmological viability (quintessence, $w_Q \rightarrow -1$).



Thank You

With thanks also to my silent AI partners

OpenAI Prism

Google Gemini

Microsoft Copilot